TUSHAR BHARADWAJ

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PROFESSIONAL SUMMARY

Remote Sensing and GIS Lead with 7+ years of experience applying ML, Computer Vision, and geospatial AI to real-world climate and carbon intelligence challenges. Proven expertise in end-to-end ML system design, from problem framing and feature engineering to scalable deployment. Strong background in geospatial ML, satellite imagery (optical, SAR, LiDAR) and carbon MRV systems (REDD+, ARR, IFM). Currently leading planning, critical thinking, and productization of ML driven GIS based carbon tools at Thryve Earth for external stakeholders and enterprise adoption.

CORE SKILLS

Machine Learning & AI: Applied ML, Deep Learning, Computer Vision, Feature Engineering, Model Validation

Production ML & MLOps: ML pipelines, scalable deployment, monitoring, cloud-native systems
Geospatial & Remote Sensing: Satellite imagery, SAR, LiDAR, time-series analysis Carbon & Climate Tech: Biomass modeling, MRV systems, REDD+, ARR, EUDR
Leadership: ML roadmap definition, cross-functional collaboration, product thinking

WORK EXPERIENCE

Thryve Earth - Remote Sensing and GIS Lead (Contract)

Mar 2024 – Present

- Own ML strategy, planning, and productization of carbon and geospatial intelligence tools designed for external commercialization.
- Automation of Carbon modeling for various project types.
- Architect scalable ML systems for biomass estimation, land-use change, and carbon accounting aligned with VCS & Gold Standard.
- Develop and deploy ML/CV models for forest cover change, degradation detection, and ARR/REDD+ prospecting.
- Lead automation of high-throughput satellite data pipelines using Python, GEE, and GCP.

- Work closely with product, engineering, and carbon teams to embed ML into core platforms.
- Translate ambiguous business and carbon methodology requirements into production-ready ML solutions.

Plant-for-the-Planet - Remote Sensing Data Scientist (Part-time)

Sep 2023 – Present

- End-to-end ownership of ML and geospatial intelligence across the organization.
- Led development of the Tropical Forest Forever Facility (TFFF) Watch Tool from concept to deployment.
- Built ML-driven biomass monitoring and time-series change detection pipelines.
- Designed EUDR compliance systems integrating land-use, deforestation, and commodity datasets.
- Developed automated tools for fire, flood, and afforestation monitoring.

IIT Roorkee - Research Fellow (ML & Computer Vision)

Apr 2022 – Mar 2024

- Built ML and CV pipelines for urban forestry using multispectral and LiDAR data.
- Developed tree canopy extraction and biomass estimation models.
- Led model experimentation, validation, and benchmarking.
- Integrated ML workflows with cloud-based remote sensing ETL pipelines.
- Created interactive dashboards using GEE and JavaScript.

Govt. College of Technology and Engineering - Assistant Professor (GIS & RS)

September 2019 – February 2022

- Taught ML-driven geospatial analytics, Python-based data analysis, and GIS.
- Supervised applied ML and remote sensing projects.
- Mentored students on real-world data science problem-solving.

Amar Found – Senior Geotechnical Engineer

May 2019 – September 2019

- Executed projects involving diaphragm walls, soil anchors, and topographic surveying.

Lovely Professional University – Assistant Professor

August 2017 – April 2019

- Delivered courses in Civil Engineering and supervised research on soil stabilization using agri-waste.

- Contributed to syllabus development and curriculum modernization.

EDUCATION

- GIS and Remote Sensing Courses, IIT Roorkee (9.5/10)
- Post Graduate Diploma in Data Science, IIIT Bangalore (3.12/4.0)
- M.Tech – Geotechnical Engineering, NIT Hamirpur (7.6/10)
- B.Tech – Civil Engineering, Rajasthan Technical University (64.8%)

TECHNOLOGY STACK

ML & AI: Python, Scikit-learn, TensorFlow, OpenCV, CNNs, RF, SVM, Time-Series
Cloud & MLOps: GCP, BigQuery, Cloud Storage, ML Pipelines
Geospatial: Google Earth Engine, GDAL, Rasterio, GeoJSON, QGIS, ArcGIS
Carbon & Climate: Biomass modeling, MRV workflows, REDD+, ARR, EUDR

PROJECT HIGHLIGHTS

- **Carbon Project Prospecting Platform** – ML-based ARR/REDD+ suitability analysis
- **Global Biomass Estimation Model** – GEDI + SAR + optical ML pipeline
- **EUDR Compliance Tool (Tracer)** – Production geospatial ML web application
- **Urban Tree Species Mapping** – CV-based classification system

PUBLICATIONS

Bharadwaj, T., Kumar, A., & Khare, S. (2023). Smart agriculture platforms: Case studies. In *Technological Prospects and Social Applications of Society 5.0* (pp. 169–176). CRC Press. https://doi.org/10.1201/9781003324720-15

Khare, S., Bharadwaj, T., Wood, S. L. R., Theron, J., Filotas, E., & Gonzalez, A. (2023). "Machine learning and satellite-based multi-scale segregation of coniferous and deciduous trees: a case study for Montreal city, Canada." GEO BON Global Conference: Monitoring Biodiversity for Action.

LANGUAGES

English (Fluent), Hindi (Fluent)

PERSONAL DETAILS

Date of Birth: 19 Sept 1994
Nationality: Indian